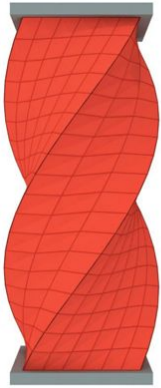


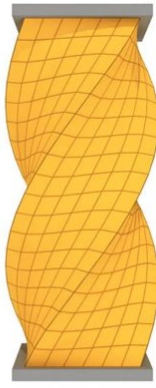
High-order elements in Position Based Dynamics

Bastien Saillant, Florence Zara, Fabrice Jaillet, Guillaume Damiand

INSA Lyon, CNRS, Universite Claude Bernard Lyon 1, LIRIS, UMR5205, 69621 Villeurbanne, France



Pyramid



Prism



Hexa (Q1)



Tetra (P1)



Tetra (P2)



Tetra (P3)

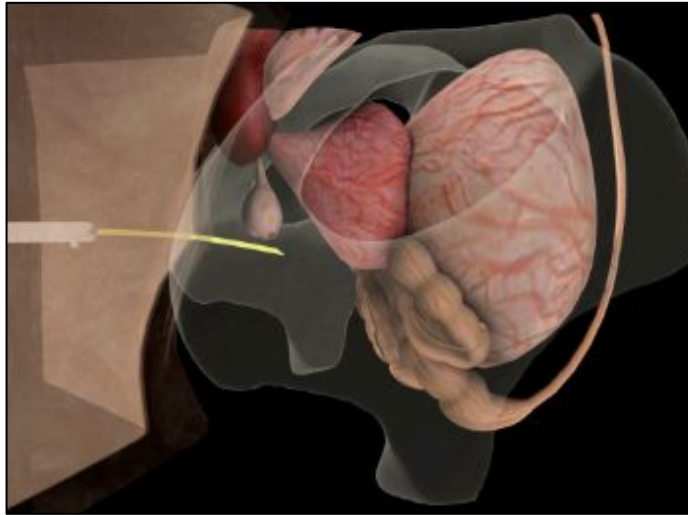
Presenter : Bastien Saillant

Problematic and Contributions

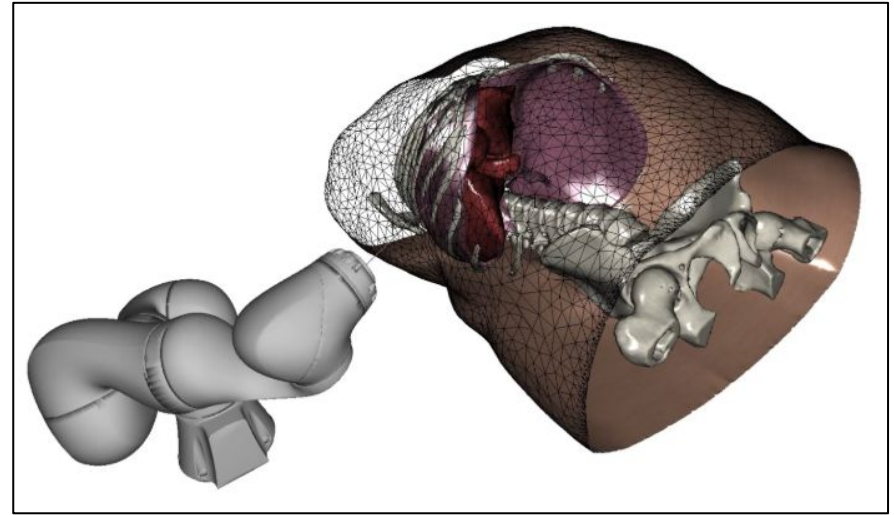
Methods

Results

Conclusion and Limitations



Interactive Simulation of Surgical Needle Insertion and Steering
Chentanez et al 2009



Interactive Finite Element model of needle insertion and laceration
Perrusi et al 2021

Needs fast and accurate simulation

Complex materials

Complex shapes

eXtended Position Based Dynamic

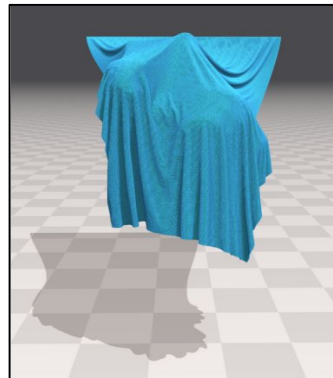
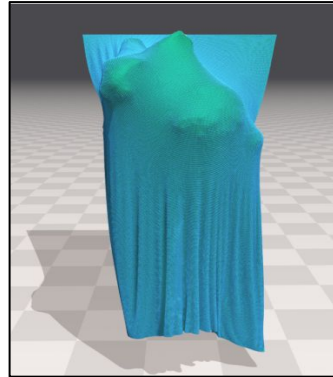
- Fast, Robust and Simple
- Local Solver (no system assembly)
- Does not require hessian
- No locking artifacts

Hyperelasticity

- large deformation
- incompressible material (ex: Neo-Hookean)
- inversion robustness
- rest stability

High-Order elements

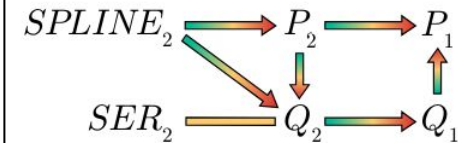
- Less topology dependent
- Better precision than linear elements
- No locking artifacts



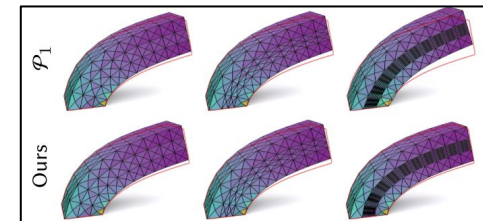
[Macklin 2019]



[Smith 2018]



[Schneider 2022]



[Schneider 2018]

[Macklin 2019] - Small Steps in Physics Simulation

[Smith 2018] - Stable Neo-Hookean Flesh Simulation

[Schneider 2018] - Decoupling Simulation Accuracy from Mesh Quality

[Schneider 2022] - A Large-Scale Comparison of Tetrahedral and Hexahedral Elements for Solving Elliptic PDEs with the Finite Element Method

Challenges (XPBD) :

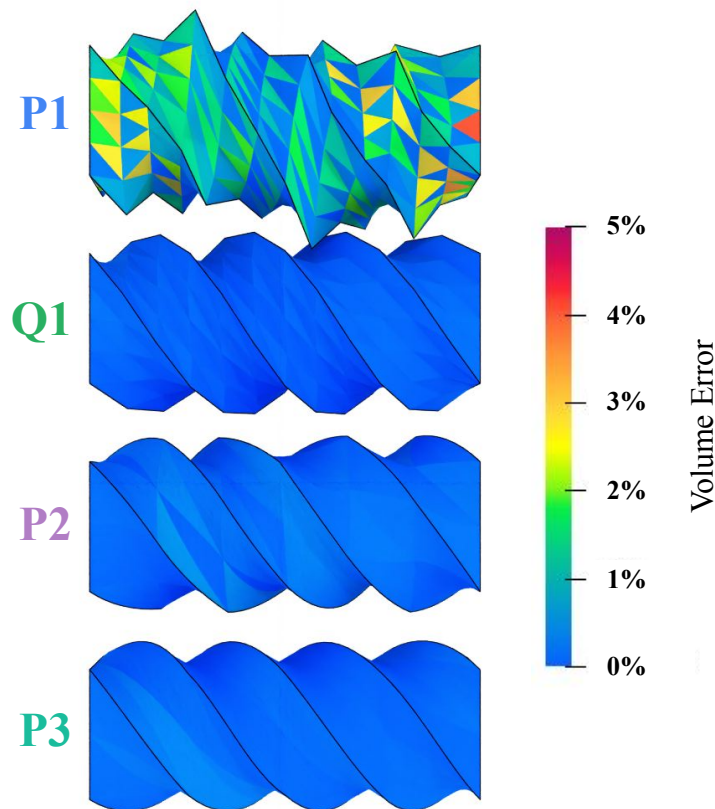
- Fast physical based [Macklin et al 2016]
- Improved convergence [Macklin et al 2019]
- Incompressible materials [Macklin et al 2021]

Limitations :

- Linear tetrahedron only (topology dependent)
- Naive incompressible materials (bad convergence)
- Can not represent very stiff material

Contributions :

- Improved accuracy with high-order elements
- Better convergence for incompressible materials



$$\nu = 0.499$$

Problematic and Contributions

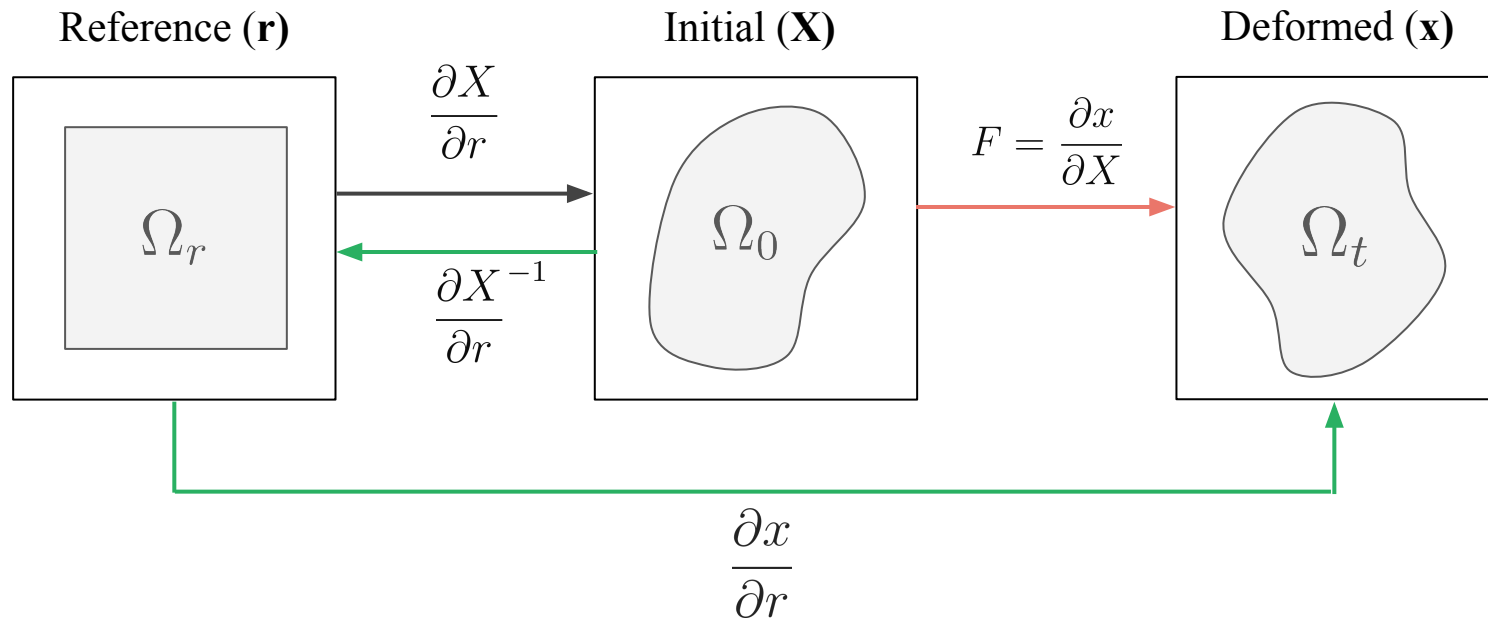
Methods

Results

Conclusion and Limitations

Deformation gradient

$$F = \frac{\partial x}{\partial r} \frac{\partial X^{-1}}{\partial r} \longrightarrow U(F) = \int_{\Omega} \Psi(F) dX \quad f(F) = - \int_{\Omega} \frac{\partial \Psi(F)}{\partial x} dX$$



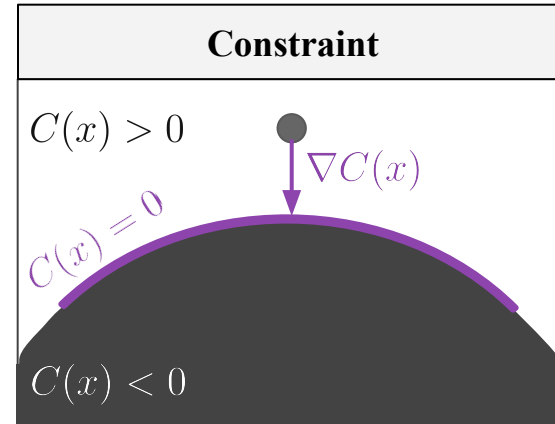
XPBD

```

predict  $\tilde{x} = x_n + \Delta t_s v_n + \Delta t_s^2 M^{-1} f_{ext}(x_n)$ 
while  $i < n_{it}$  do
  for all constraints do
    compute  $C(x)$  and  $\nabla C(x)$ 
    compute  $\Delta\lambda$ 
    compute  $\Delta x$ 
    update  $\lambda_{i+1} \leftarrow \lambda_i + \Delta\lambda$ 
    update  $x_{i+1} \leftarrow x_i + \Delta x$ 
  end for
   $i \leftarrow i + 1$ 
end while
update velocities  $v_{n+1} \leftarrow \frac{1}{\Delta t_s} (x_{n+1} - x_n)$ 
    
```

In practice : only 1 iteration

Small time steps [Macklin 2019]



↑ $C(x) ?$

Energy and Forces

$$U(F) = \int_{\Omega} \Psi(F) dX \quad f(F) = - \int_{\Omega} \frac{\partial \Psi(F)}{\partial x} dX$$

$$U(F) = \int_{\Omega} \Psi(F) dX \quad f(F) = - \int_{\Omega} \frac{\partial \Psi(F)}{\partial x} dX$$

Energy

$$U(F) = \frac{1}{2} \alpha^{-1} C(F)^2 \quad f(F) = -\nabla C(F) \alpha^{-1} C(F) \quad \alpha = k^{-1}$$

Energy as constraints

[Macklin 2016]

$$C(F) = \left(\int_{\Omega} \tilde{\Psi}(F) dX \right)^{0.5} \quad \nabla C(F) = \frac{1}{2C(F)} \int_{\Omega} \frac{\partial \tilde{\Psi}(F)}{\partial x} dX \quad \tilde{\Psi}(x) = 2\alpha \Psi(x)$$

Constraints

$$C(F) = \left(\sum_i^n V_i \tilde{\Psi}(F_i) \right)^{0.5} \quad \nabla C(F) = \frac{1}{2C(F)} \sum_i^n V_i \frac{\partial \tilde{\Psi}(F_i)}{\partial x} \quad V_i = w_i |det(J_i(X))|$$

Isoparametric Elements
+ Quadrature

Constraint has only 1 stiffness parameter

Energy = N × Constraint

Rest stable energy: $F = \mathbb{I} \quad \frac{\partial \Psi}{\partial F} = 0$

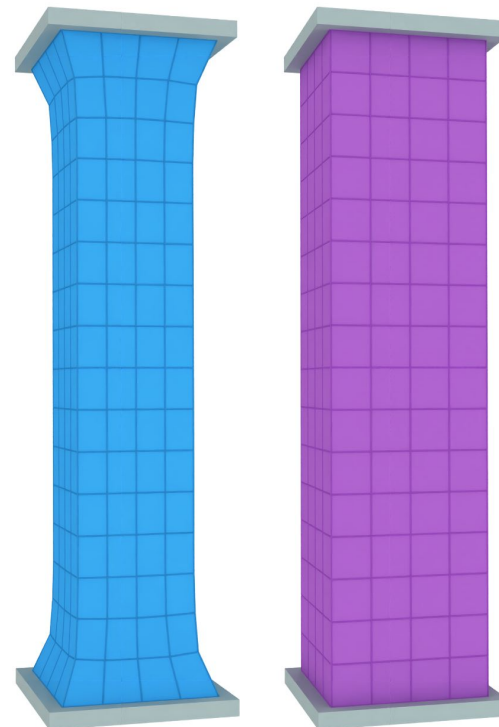
Unstable	$\Psi_{NH} = \frac{\mu}{2}(tr(F^T F) - 3) + \frac{\lambda}{2}(I_3 - 1)^2 \quad I_3 = det(F)$
----------	--

$$\frac{\partial \Psi_{NH}}{\partial F} = \mu F + \lambda(I_3 - 1) \frac{\partial I_3}{\partial F} \quad \frac{\partial \Psi_{NH}}{\partial F}(\mathbb{I}) = \boxed{\mu \mathbb{I}} \text{ forces at rest !}$$

Stable	$\Psi_{SNH} = \frac{\mu}{2}(tr(F^T F) - 3) + \frac{\lambda}{2}(I_3 - \gamma)^2 \quad \gamma = 1 + \frac{\mu}{\lambda}$
--------	--

$$\frac{\partial \Psi_{SNH}}{\partial F} = \mu F + \lambda(I_3 - \gamma) \frac{\partial I_3}{\partial F} \quad \frac{\partial \Psi_{SNH}}{\partial F}(\mathbb{I}) = \boxed{\mu \mathbb{I} + \lambda(1 - \gamma) \mathbb{I} = 0}$$

stable



Rest stable constraint: $F = \mathbb{I} \quad \frac{\partial \tilde{\Psi}}{\partial F} = 0 \quad \tilde{\Psi} = 2\alpha\Psi$

$$\Psi_{SNH} = \frac{\mu}{2}(tr(F^T F) - 3) + \frac{\lambda}{2}(I_3 - \gamma)^2 \quad \gamma = 1 + \frac{\mu}{\lambda}$$

Unstable

$$\tilde{\Psi}_A = tr(F^T F) - 3$$

$$\tilde{\Psi}_B = (I_3 - \gamma)^2$$

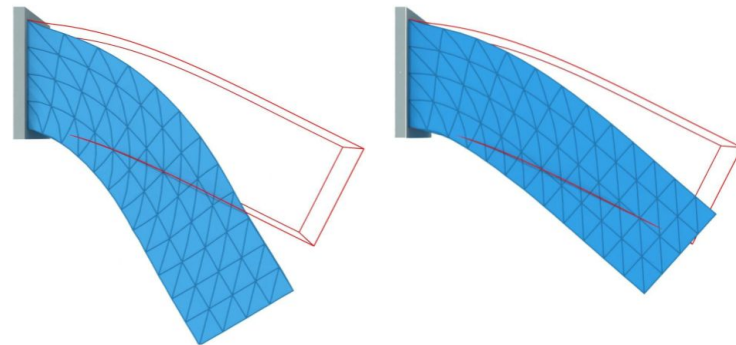
A stable energy does not guaranty stable constraints !

$$\Psi_{SNH} = \frac{\mu}{2}(tr(F^T F) - 2I_3 - 1) + \frac{\lambda}{2}(I_3 - 1)^2$$

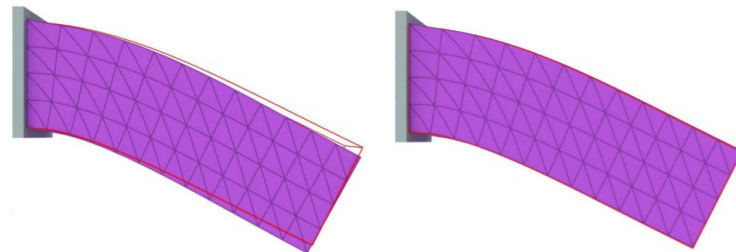
Stable

$$\tilde{\Psi}_A = tr(F^T F) - 2I_3 - 1$$

$$\tilde{\Psi}_B = (I_3 - 1)^2$$



20 sub-iterations



50 sub-iterations

Problematic and Contributions

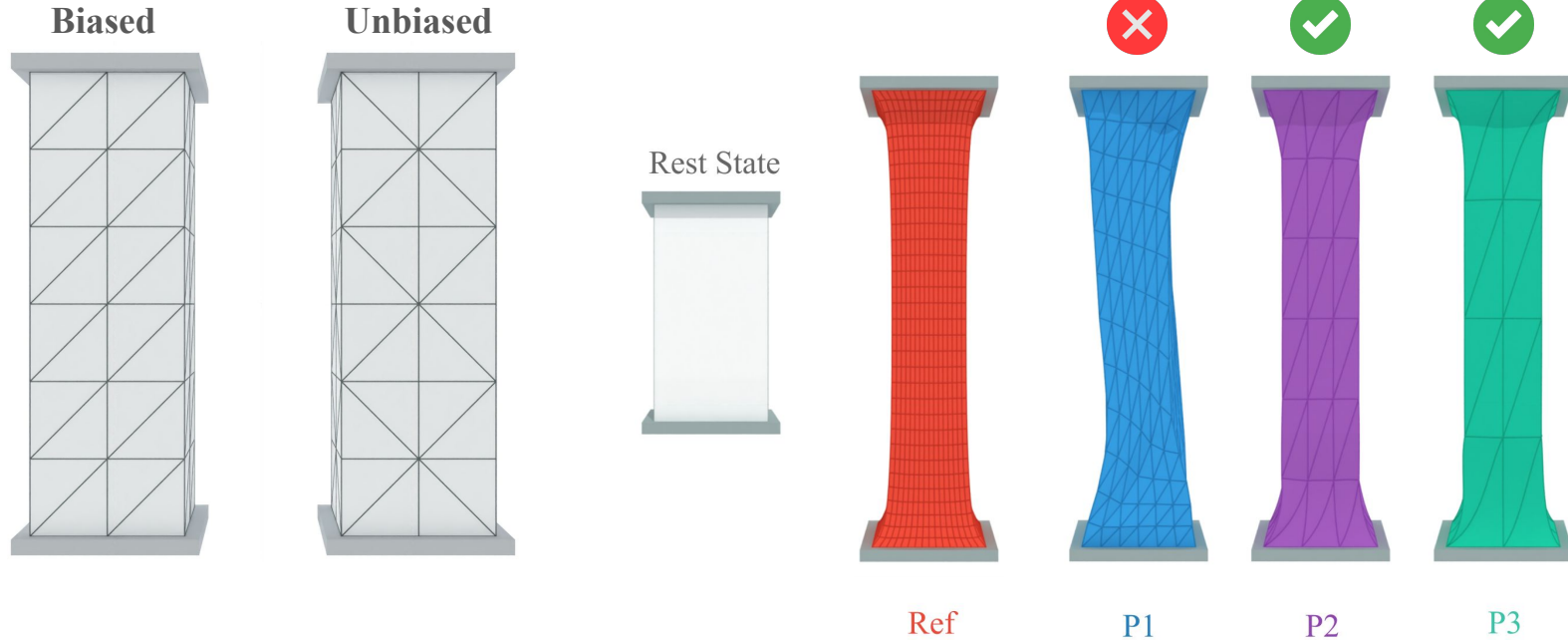
Methods

Results

Conclusion and Limitations

Result: Mechanical Tests (Stretch)

Is it accurate given a bad topology and high-volume preservation ($v = 0.499$)?

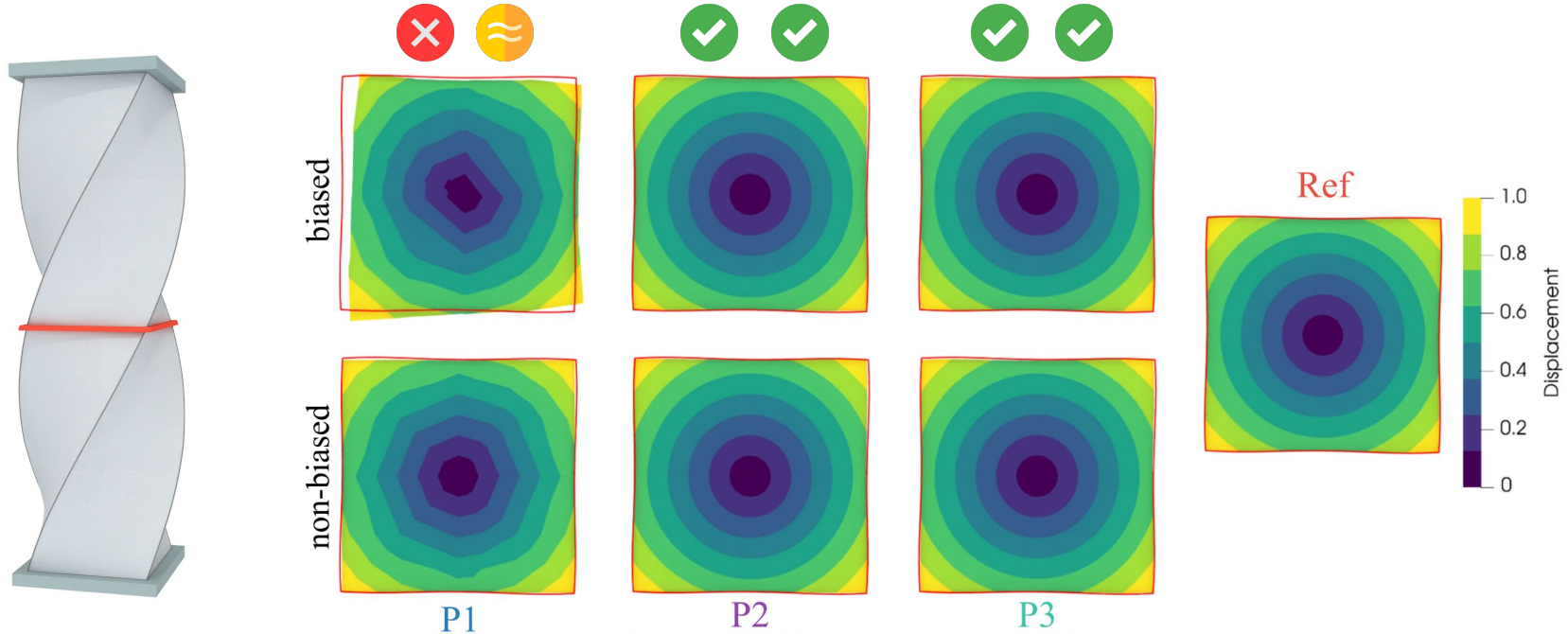


P1 elements suffer from biased mesh

Coherent behaviour for P2 and P3 elements

Result: Mechanical Tests (Twist)

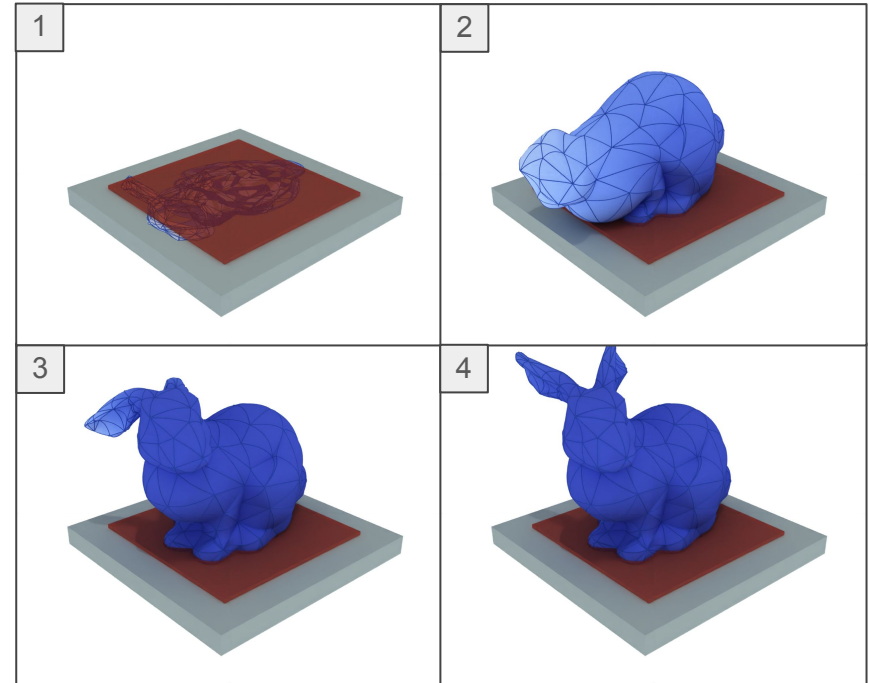
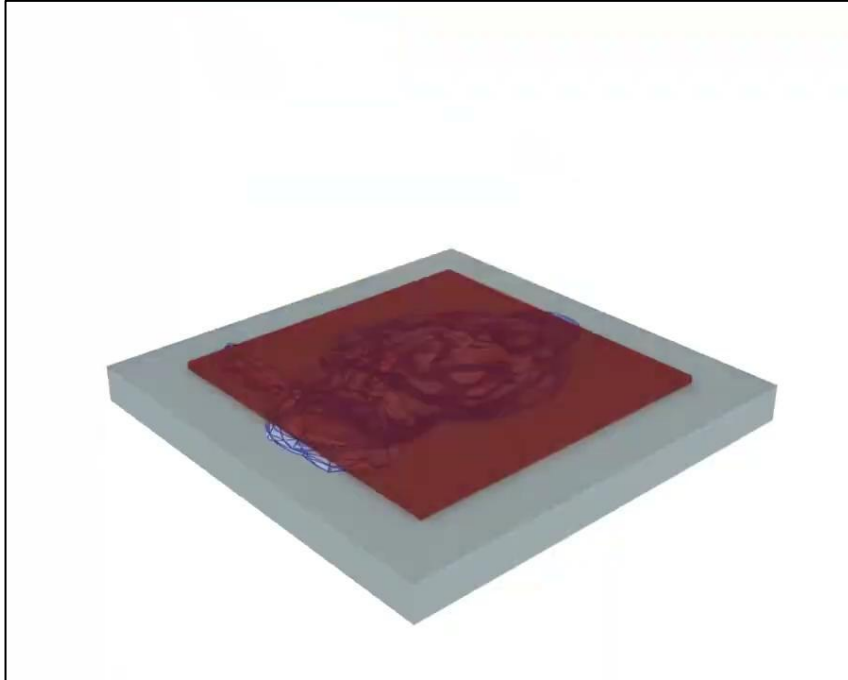
With low poisson coefficient ($\nu = 0.35$) and non-biased mesh?

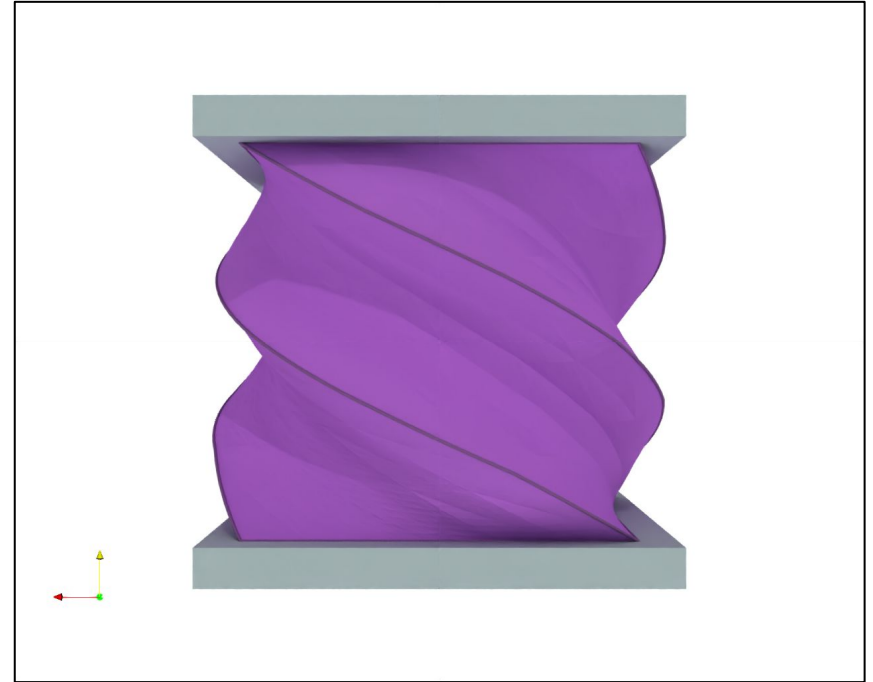
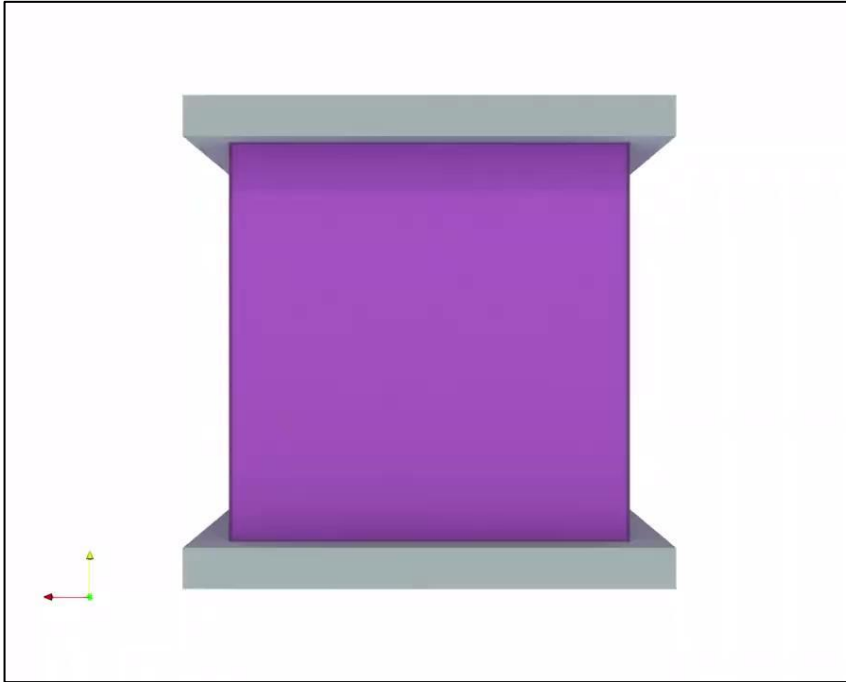


P1 are sensible to discretization and rough solution

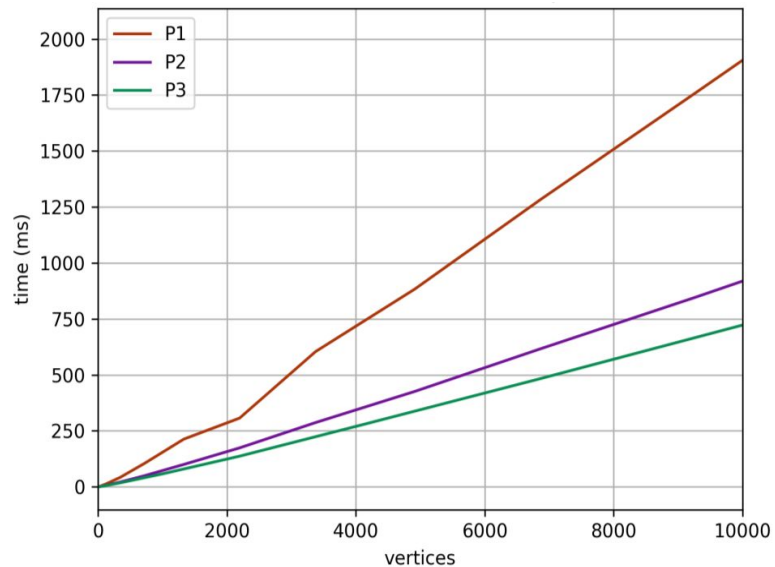
Biased or not P2/P3 are close to the reference solution

Result: Stability

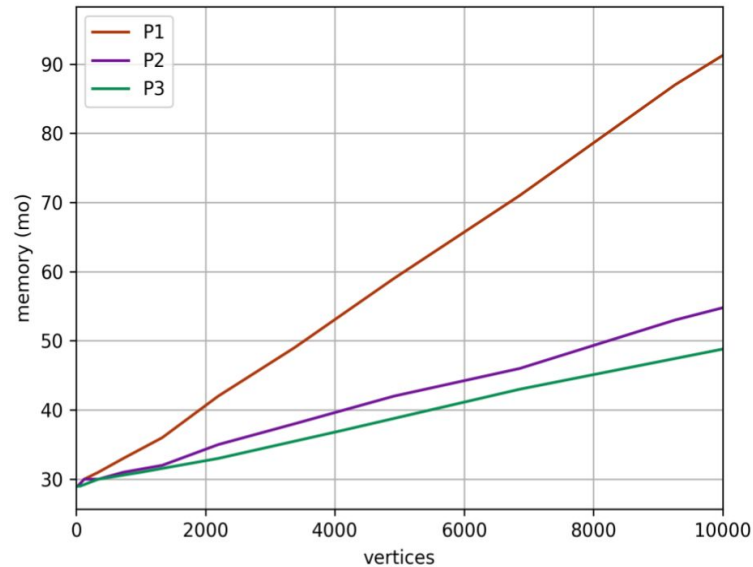




Simulation loop cost (ms)



Memory (mo)



First derivative → increase linearly

Faster and less memory for the same number of vertices

Result: Performance

Initial Mesh



#vertices
#elements

P1 Coarse
(30 ms)



356
1 279

P1 Refined
(2184 ms)



15 799
81 793

P2
(84 ms)



2 252
1 279

P3
(314 ms)



6967
1 279

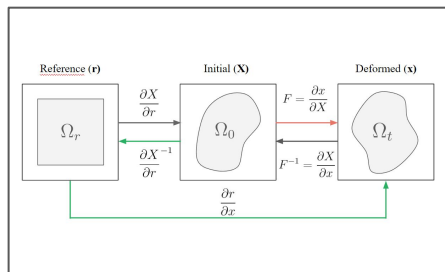
Problematic and Contributions

Methods

Results

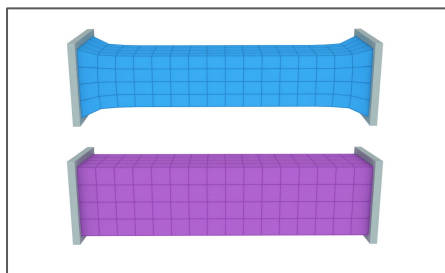
Conclusion and Limitations

Isoparametric elements

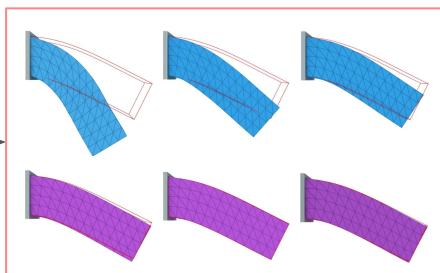


Constraint formulation

$$\begin{aligned}
 U(x) &= \int_{\Omega} \Psi(x) dX & f(x) &= - \int_{\Omega} \frac{\partial \Psi(x)}{\partial x} dX \\
 U(x) &= \frac{1}{2} \alpha^{-1} C(x)^2 & f(x) &= - \nabla C(x) \alpha^{-1} C(x) & \alpha &= k^{-1} \\
 C(x) &= \int_{\Omega} \tilde{\Psi}(x) dX & \nabla C(x) &= \frac{1}{2C(x)} \int_{\Omega} \frac{\partial \tilde{\Psi}(x)}{\partial x} dX & \tilde{\Psi}(x) &= 2\alpha \Psi(x) \\
 C(x) &= \sum_i^n V_i \tilde{\Psi}(F_i) & \nabla C(x) &= \frac{1}{2C(x)} \sum_i^n \frac{\partial \tilde{\Psi}(F_i)}{\partial x} & V_i &= w_i \det(J_i(X))
 \end{aligned}$$

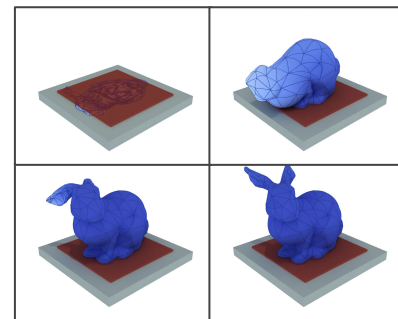
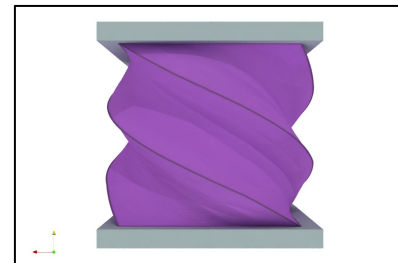


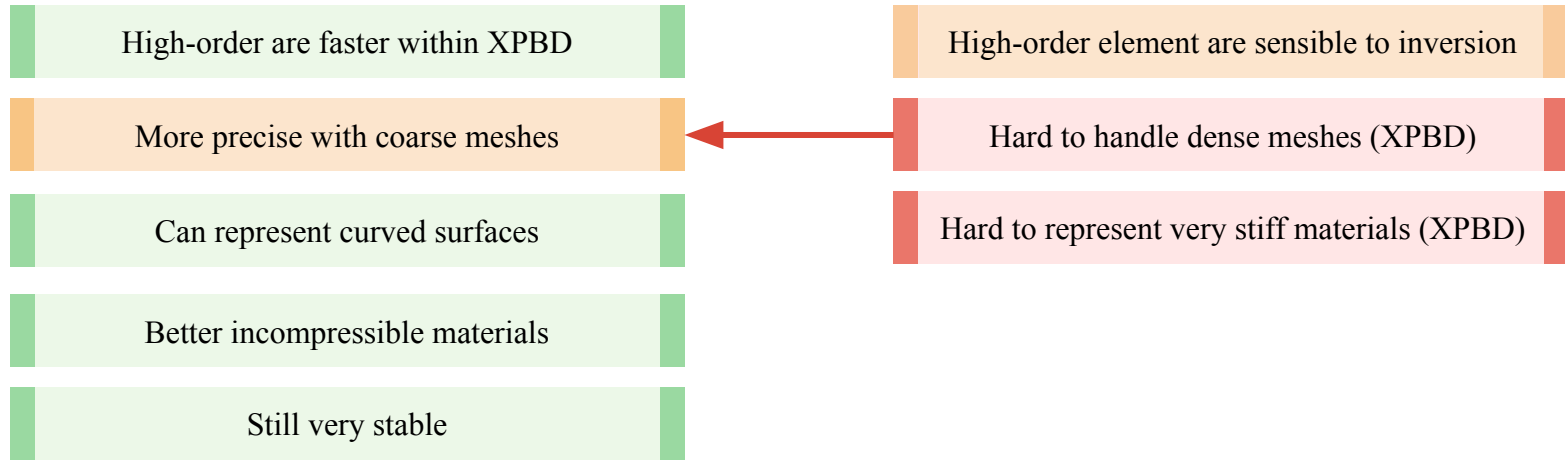
Rest stable energies



Rest stable constraints

XPBD





Perspectives





Thank You!

Contact: bastien.saillant@liris.cnrs.fr



Position Based Dynamic

XPBD

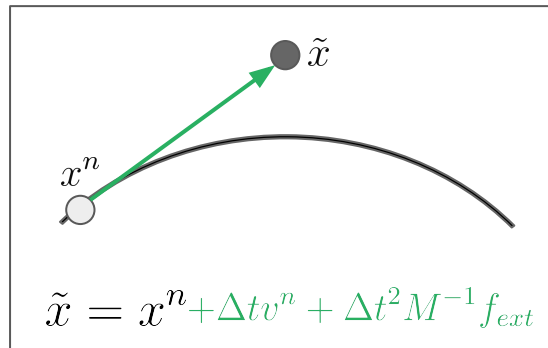
```

predict  $\tilde{x} = x_n + \Delta t_s v_n + \Delta t_s^2 M^{-1} f_{ext}(x_n)$ 
while  $i < n_{it}$  do
  for all constraints do
    compute  $C(x)$  and  $\nabla C(x)$ 
    compute  $\Delta\lambda$ 
    compute  $\Delta x$ 
    update  $\lambda_{i+1} \leftarrow \lambda_i + \Delta\lambda$ 
    update  $x_{i+1} \leftarrow x_i + \Delta x$ 
  end for
   $i \leftarrow i + 1$ 
end while
update velocities  $v_{n+1} \leftarrow \frac{1}{\Delta t_s}(x_{n+1} - x_n)$ 
  
```

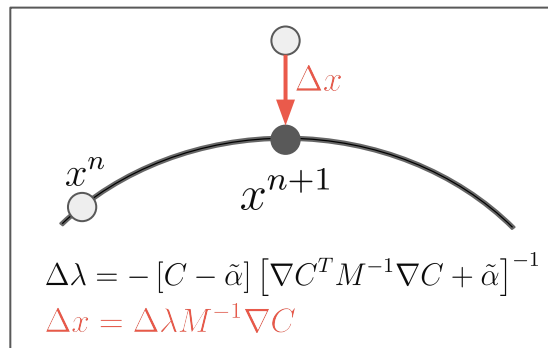
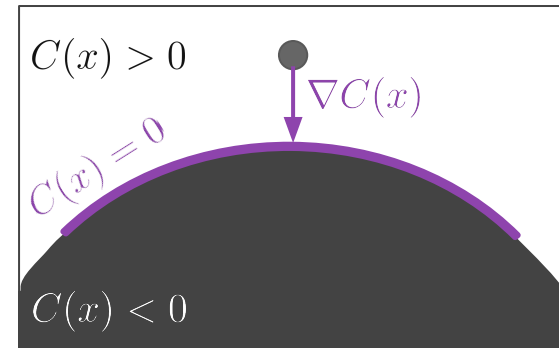
In practice : only 1 iteration

Small time steps [Macklin et al 2019]

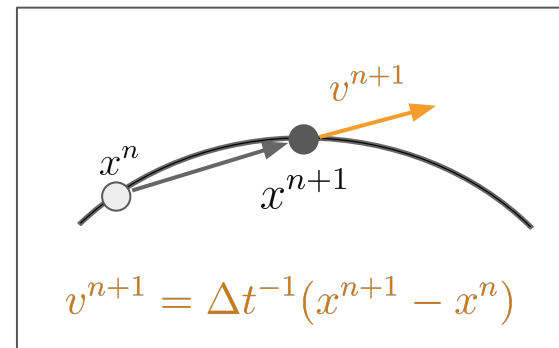
First Guess



Evaluate Constraint



Projection



Update Velocity